

Topology First-Year Exam, May 2019, Part 2

Answer all questions and work all problems. Make proofs as succinct as possible.

1. Let $n \geq 1$. Suppose that $f, g : X \rightarrow \mathbb{R}^n$ are continuous. Show that f and g are homotopic.
2. Let $\gamma : [0, 1] \rightarrow X$ with basepoint x_0 . Let γ^{-1} be defined by $\gamma^{-1}(t) = \gamma(1 - t)$. Show that $\gamma * \gamma^{-1}$ is loop homotopic to the constant loop $1 : [0, 1] \rightarrow X$, $1(t) \equiv x_0$.
3. Show that $\pi_1(S^1, 1) = \mathbb{Z}$.
4. Show that the disk, $D^2 = \{z \in \mathbb{C} \mid \|z\| \leq 1\}$, has the fixed point property.
5. Consider the matrix $M = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$. This represents a group homomorphism $M : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$. Show that there is a map $f : T^2 \rightarrow T^2$ such that $f_* = M$, where $f_* : \pi_1(T^2) \rightarrow \pi_1(T^2)$ is the homomorphism induced by f on the fundamental group.
6. Suppose that $f : S^1 \rightarrow S^1$ is continuous. Show that f is homotopic to $z^n : S^1 \rightarrow S^1$ for some $n \in \mathbb{Z}$. This is the degree of f .
7. Let $n > 1$. Let $\mathbb{P}^n = S^n/D$, where D identifies x with its antipode for all $x \in S^n$. Show that $\pi_1(\mathbb{P}^n, x_0) \cong \mathbb{Z}_2$.
8. Describe a space X such that $\pi_1(X, x_0) \cong \mathbb{Z}_3$. Verify that this is the fundamental group by the Seifert–van Kampen Theorem.
9. A topological space is said to be Lindelöf provided that for every open cover $\mathcal{U}$ of X , there is a countable $\mathcal{V} \subset \mathcal{U}$ covering X . Show that if X is a separable metric space, then X is Lindelöf.
10. State and prove the Tychonoff Theorem.
11. State the following theorems or verify the following statements with quick examples or statements.
 - The Alexander Subbase Theorem.
 - The Hahn–Mazurkiewicz Theorem
 - The Brouwer Characterization of the Cantor Set
 - The Jordan Curve Theorem
 - The Borsuk–Ulam Theorem for S^2 .
 - Determine the fundamental group of T^2 .
 - Determine the fundamental group of the connected sum of two tori, $\pi_1(T^2 \# T^2)$.
 - The Arcwise Connectedness Theorem
 - Suppose that X is a compact metric AR. Show that for any Z and any pair of continuous functions $f, g : Z \rightarrow X$, f and g are homotopic.
 - Suppose that C is the Cantor set. Show that C is homeomorphic to $C \times C$.